

Macroeconomics #0

Introduction

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About Me

- Name: Tran Quang-Thanh
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- Current position:
 - Visiting Fellow at Research Center for Policy Design, Tohoku University
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- Research Interests: macroeconomics, economic growth, life-cycle dynamics, demographic economics, and monetary economics.
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Why should you take this course?

- Bachelor economics in general lacks theoretical framework. In recent years, many universities in US, Europe, Japan, and China have introduced micro-foundation and mathematical tools to 3rd and 4th year students.
- As far as I know, very few Vietnam's graduate schools offer adequate education and training on modern macroeconomics with micro foundations, structural estimation, multi-sectoral monetary economics, etc.
- Although many of the materials will be covered in your future graduate school, this course will be great for a transition.

Some short history

Mainstream

- ① Classical OLG and neoclassical growth: Solow, Samuelson, Diamond (1950s–1960s)
- ② Lucas critique and new classical revolution (1970s)
- ③ RBC and quantitative DSGE (1980s)
- ④ New Keynesian DSGE and monetary policy (1990s–2000s)
- ⑤ Frictions, crises, and limits of RANK (1990s–2010s)
- ⑥ Heterogeneous agents and HANK (2010s–today)

Growth and Life-Cycle branches

- Endogenous growth: Romer, Jones, Aghion–Howitt (1990s)
- Quantitative OLG: Auerbach–Kotlikoff, Greenwood (1980s–1990s)
- Demographics, human capital, unified growth: Galor, de la Croix (2000s–2010s)

What will we study?

- ① Basic mathematical tools
- ② Solow Model
- ③ Ramsey Model
- ④ OLG Model
- ⑤ Some Two-period Models
- ⑥ Endogenous Growth Model
- ⑦ Real Business Cycle Model

Basically, we will go till the 1990s :)

Some books that will help you along the way

Math and dynamical analysis

- Sydsaeter, K., Seierstad, A. & Strom, A. (2008). Further mathematics for economic analysis. Pearson education.
- Simon, C. P., & Blume, L. (1994). Mathematics for economists (Vol. 7). New York: Norton.
- Sorger, G. (2015). Dynamic Economic Analysis. Cambridge University Press.
- Shone, R. (2002). Economic Dynamics: Phase diagrams and their economic application. Cambridge University Press.

Dynamic Programming and Optimal Control

- Stokey, N. L., & Lucas Jr, R. E. (1989). Recursive methods in economic dynamics. Harvard University Press.
- Ljungqvist, L., & Sargent, T. J. (2018). Recursive macroeconomic theory. MIT press.
- Weber, T. A. (2011). Optimal control theory with applications in economics. MIT press.
- Kamien, M. I., & Schwartz, N. L. (2012). Dynamic optimization: the calculus of variations and optimal control in economics and management. courier corporation.

Suggested Books for This Course

- Chu, A. C. (2021). Advanced Macroeconomics: An introduction for undergraduates.
- Jones, C., & Vollrath, D. (2013). Introduction to Economic Growth. 3rd ed.
- Acemoglu, D. (2008). Introduction to modern economic growth. Princeton university press.
- Barro, R., & Sala-i-Martin, X. (2004). Economic growth second edition.
- Campante, F., Sturzenegger, F., & Velasco, A. (2021). Advanced macroeconomics: an easy guide. LSE Press.
- Romer, D. (2019). Advanced Macroeconomics. 5th Ed.
- de la Croix, D., & Michel, P. (2002). A theory of economic growth: dynamics and policy in overlapping generations. Cambridge University Press.
- McCandless, G. (2009). The ABCs of RBCs: An introduction to dynamic macroeconomic models. Harvard University Press.